11th grade - Learning evidence 2.1: Confidence Interval Planning

Name:____

Assessment and evaluation

This exam evaluates only the criteria C1–C5. Each criterion is evaluated across the three problems below. A criterion is passed if it is correctly activated in at least two of the three problems. The overall exam result is binary (Passed/Not passed).

Problem 1

A city budgeting office wants to estimate the average weekly transit subsidy per rider for two pilot zones. Zone A recorded the following sample of weekly subsidy amounts (USD):

- Zone A sample ($n_A = 6$): 18, 22, 19, 21, 20, 17.
- Zone B sample ($n_B = 8$): 24, 26, 23, 25, 27, 22, 24, 26.

An annual audit provides the population standard deviations for the zones: $\sigma_A = 3{,}5$ and $\sigma_B = 4{,}0$.

(C1) Compute the sample mean and sample standard deviation for each zone.

(C2) Distinguish between the sample statistics and the population parameters given in the description.

(C3) Explain why the office should use interval estimation instead of a single point estimate for budgeting.

(C4) Interpret the meaning of a 95 % confidence interval for the mean weekly subsidy in each zone.

(C5) Construct 95 % confidence intervals for the mean weekly subsidy in Zone A and Zone B using the known population standard deviations.

Problem 2

A regional warehouse studies the number of orders processed per hour to plan staffing. A random sample of 60 hours is summarized in grouped form below. The operations team also reports a known population standard deviation of $\sigma = 6{,}2$ orders per hour.

Orders per hour (x)	Frequency (f)
14	8
18	16
22	20
26	10
30	6

(C1) Compute the grouped sample mean and sample standard deviation of orders per hour.

(C2) Identify which values represent sample statistics and which represent population parameters.

(C3) Explain the difference between reporting a point estimate and reporting a confidence interval in this setting.

(C4) Interpret a 95 % confidence interval for the mean hourly order rate in terms of staffing decisions.

(C5) Construct a 95 % confidence interval for the mean hourly order rate using the known population standard deviation.

Problem 3

A logistics firm monitors the time (in minutes) required to load containers at a port. A sample of 80 loading times is grouped into class intervals below. The firm has a historical estimate of the population standard deviation, $\sigma = 5{,}8$ minutes.

Loading time interval (minutes)	Frequency (f)
20–24	12
25–29	18
30–34	20
35–39	17
40–44	13

(C1) Use class midpoints to compute the grouped sample mean and sample standard deviation.

(C2) Distinguish the sample statistics from the population parameter supplied in the description.

(C3) Explain why an interval estimate is more informative than a single mean when negotiating service-level targets.

(C4) Interpret a 95 % confidence interval for the mean loading time in the context of port scheduling.

(C5) Construct a 95 % confidence interval for the mean loading time using the known population standard deviation.